

2.5 RESEARCH ASSUMPTIONS

- Market values have been rounded off to the nearest decimal. Due to this, there may be slight differences between actual totals and the totals mentioned in market tables.
- Region-wise inflation/deflation for service pricing during 2023–2028 has not been considered.
- It has been assumed that dollar fluctuations will not affect the growth forecast to a significant extent.
- Revenues of private companies acquired from paid databases were assumed to be in line with actual company financials.
- A positive economic climate will continue in emerging countries until 2028, and the recession of 2023 will have a low impact on the microbiome sequencing services market.
- There will be no substantial changes in government regulations that will significantly impact the market.
- All market segments and subsegments listed in this report are considered to be mutually exclusive.
- Wherever the exact segmental market splits were not available, historical trends and inputs from primary respondents have been considered.

2.6 RISK ANALYSIS

DESCRIPTION	RISK ANALYSIS
The growth rate for each segment is considered based on recent market trends, key developments of companies, analysis of historical growth rates, and primary insights.	Moderate
Market sizes and shares for 2022 were estimated based on the financials available for nine months in company annual/quarterly reports and presentations for a few companies.	Low to Moderate

2.7 IMPACT OF RECESSION ON MICROBIOME SEQUENCING SERVICES MARKET

A recession is a period of temporary economic decline during which trade and industrial activities are reduced. It is generally identified by a decline in the GDP in two successive quarters. According to the World Economic Forum Forecast Report (October 2022), global GDP growth would slow from 6% in 2021 to 3.2% in 2022 and 2.7% in 2023. Alternatively, the same report suggests that global inflation would rise from 4.7% in 2021 to 8.8% in 2022 but decline to 6.5% in 2023 and to 4.1% by 2024.

The key reasons behind this slowdown are the Russia-Ukraine war (which led to a severe energy crisis in Europe, where gas prices have escalated four-fold since 2021) and frequent lockdowns in China under its 'Zero COVID policy,' which impacted global trade and led to a sharp appreciation of the US dollar.

Based on analysis by the pharmaceutical and biotechnology industry and inputs from primary experts, the pharmaceutical and biotechnology industry is expected to withstand this downturn. The demand for healthcare products and services is expected to endure, and the recession will have a low to minimal impact on overall healthcare-oriented businesses.

Healthcare, inclusive of pharma-biotech companies, is a defensive industry, which is likely to hold up relatively well to recession situations compared to other industries such as automotive, telecom, and energy, as drugs for treating health conditions are a necessity for patients. The industry is also backed by insurance and health plans, which make this sector recession-resilient.

Additionally, primary experts mentioned that investors are now looking to invest in defensive businesses (as a hedge against recessionary views), such as the pharmaceutical and biotechnology industry, and would avoid recession-sensitive businesses like consumer durables. On the other hand, small to medium-sized pharmaceutical and biotechnology companies might face uncertainties due to changes in macroeconomic conditions, such as a decrease in healthcare spending, increasing cost of raw materials, reduced margins due to rising inflation, changes to the government funding environment, and limited or significantly reduced points of access of products. Such factors might have an insignificant effect on the growth of the market in the forecast period.

As per the Healthcare Outlook 2023 report published by The Economist Intelligence Unit Limited (EIU), total healthcare spending (public and private) was expected to rise by 4.9% in 2023, propelled by higher costs and wages. However, due to high inflation and slow economic growth, EIU estimated that healthcare spending would remain similar to 2022 throughout 2023, meaning that 2023 will be the second successive year of real-term funding declines. OECD data suggests that after the global financial crisis of 2008–09, healthcare spending on preventive care and pharmaceuticals was reduced, and the same pattern is expected to be repeated. This decline in healthcare spending will be partially offset by growth in the pharmaceutical industry, the expansion of international companies into developing geographies, and investments made in 2021 and 2022.

TABLE 1 GLOBAL INFLATION RATE PROJECTIONS, 2021–2027 (% GROWTH)

Inflation Rate	2021	2022	2023	2024	2025	2026	2027
	4.7%	8.8%	6.5%	4.1%	3.2%	3.3%	3.2%

Source: IMF, World Economic Outlook Update, July 2022

TABLE 2 US HEALTH EXPENDITURE, 2019–2022 (USD MILLION)

Expenditure Amount	2019	2020	2021	2022
Total National Health Expenditure	3,757,382.0	4,144,077.0	4,255,127.0	4,442,352.6

Source: Centers for Medicare & Medicaid Services, 2022 (estimated on the basis of CMC data).

TABLE 3 US HEALTH EXPENDITURE, 2023–2027 (USD MILLION)

Expenditure Amount	2023	2024	2025	2026	2027
Total National Health Expenditure	4,657,806.7	4,904,670.4	5,169,522.6	5,449,969.3	5,746,992.6

Source: Centers for Medicare & Medicaid Services, 2022–2027 (estimated on the basis of CMC data).

As per the Centers for Medicare & Medicaid Services data, US healthcare spending will grow at an average of 4.9% per year from 2022 to 2024 and 5.3% per year beyond that, hitting USD 6.8 trillion in 2030.

However, economists and other financial experts have stated that they do not observe any major reduction in the usage of pharma/biotech products and services in the slower-growing global economies. The demand for medicines is relatively inelastic. Moreover, many drugs are expected to go off-patent between 2025 to 2030. This is likely to boost drug innovation in the drug discovery and development field. Therefore, the microbiome sequencing services market will be one of the less-affected sectors.